

What SLEEP Was, What the Evidence Forced, and What It Became

SLEEP: Final Report — the Complete Arc, April to August 2026

Supersedes the 8 August final report; covers the full programme from the original system through the localisation repair and the four-model validation matrix

Prepared for: Prof. Debashis Guha
Director, MAIB; Chair, CRTIB
SP Jain School of Global Management

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Prepared by: Aditya Tripathi

Professor Guha,

This report replaces the 8 August final and tells the programme’s story whole, in the order the evidence arrived: the system as originally built, the failure it exhibited, the replication programme that made that failure rigorous, the re-examination that located its causes, and the validation matrix that turned it into a working, baseline-controlled result. The one-line arc: **a recall floor of 0.006 that survived every within-paradigm repair rose to 0.750 once we moved the consolidation write to where the transformer actually stores facts** — and the repaired system now beats both naive fine-tuning and an EWC baseline in its design regime, under pre-registered gates. Every number below traces to a released run file.

1. What SLEEP was

By late April, SLEEP was a complete, pre-registered implementation of biological memory consolidation in a transformer: a tagging layer that flags surprising spans by prediction error, a direct-write KV memory bank, a PRP-style resource allocator, and a sleep engine that replays selected memories into a LoRA consolidation adapter under four safety mechanisms (EWC, hard clipping, interleaving, validation rollback). Thirty-six design questions were resolved in a formalization document before any code was written.

The first campaign (Qwen2.5-7B, 200 synthetic single-exposure facts) produced the observation that named the paper: injected memories bias four-way *recognition* by +0.16 over untagged facts, yet *free-form recall* of the same facts is 0.00 — and no safety setting in a four-point sweep achieved useful recall at acceptable damage. Around it sat two more single-run observations: the internal validator certified consolidations that failed any direct test, and a three-cycle comparison suggested SLEEP preserved capability better than naive fine-tuning. Everything rested on one model and one seed.

2. What the review named

Your review of that first evidence, together with the external review, converged on one demand in eleven items: *nothing is replicated, nothing is compared properly, and the repair the diagnosis implies has not been tried*. Concretely: multiple seeds, extended horizons (10+ cycles), more models

and precisions, real baselines (RAG, EWC-only, in-context), an external validation gate, tightened biological framing, and the retrieval-aware fine-tune.

3. Hostile replication: what we did, what it found

We rebuilt the programme around those demands — five seeds per result, four models across three families (Qwen2.5-7B/1.5B, Llama-3.1-8B, Mistral-7B), ten-cycle horizons, all baselines, a two-stage validation gate, and a ground-truth verification harness that checks the injection path on real hardware before any run (it rejected two defective configurations; no invalid number ever entered the record). The findings:

	Qwen-7B	Qwen-1.5B	Llama-8B	Mistral-7B
Recall from consolidated weights (5 seeds)	0.006	0.007	0.006	0.007
Recall with the fact in context	0.760	0.700	0.987	0.927
Proxy validation false-confirmation	~97%	100%	~93%	100%

The failure was universal: every model reads and uses a fact it can see, and recalls essentially nothing consolidated into its weights; internal surprise-based validation is wrong about 93–100% of the consolidations it certifies, everywhere. The earlier three-cycle preservation claim failed replication — across nine seeded ten-cycle run-pairs the clip-induced damage plateau varies by three orders of magnitude on identical hyperparameters ($2.7 \rightarrow 1,531$), so we withdrew that claim and replaced it with the stronger methodological finding that fixed-horizon, single-run continual-learning comparisons are unreliable. Most uncomfortably: **EWC alone (DRA 0.070 at BCP 1.12) beat every full-pipeline configuration on recall-per-damage.**

4. What we added beyond the checklist

Two self-initiated additions defined this phase. The **retrieval-aware warm-up** — the repair the recognition–recall diagnosis implies — was promoted from side-item to a pre-registered extension (success bar fixed in advance: DRA 0.10–0.15 at BCP < 1.5) and tested properly: two variants, two seeds, four models. It is a replicated null: across sixteen runs recall moved by at most ± 0.008 , while recognition improved where the model survived — the dissociation reproduced under intervention. And the **plateau lottery** — the three-orders-of-magnitude seed variance above — emerged only because horizon and seed demands were run *together*; it generalises past SLEEP to how the continual-learning literature reports results.

At this point the paper was an honest, universal, well-controlled diagnosis — of a failure. Its title said “Empirical Limits.” The question we could not close: is consolidation into frozen LLMs impossible, or were we doing it wrong?

5. The re-examination: three stacked causes

Before accepting the negative result, we ran a pre-mortem on the paper: assume the conclusion is wrong — where would the error be? Re-deriving the design from first principles against the knowledge-editing literature (Geva et al.; ROME; MEMIT) exposed an assumption that had never been questioned: **the consolidation adapter’s placement was chosen by biological analogy** (attention projections, top third of layers, playing “neocortex”), while causal-tracing evidence

locates factual associations in *mid-stack MLP* modules. Two further suspects followed from the same pass: single-wording training (cloze succeeded while recall failed — the signature of string memorization rather than extractable knowledge) and sampled decoding in knowledge-presence evaluation (replaced with greedy throughout).

A five-arm isolation ladder (50 facts, matched exposure, two models, two seeds) tested the hypothesis:

Arm (Qwen2.5-7B)	Free-form recall	BCP
attention-top, single wording (original)	0.01	3.6
mid-MLP, single wording	0.03	1.2–1.8
mid-MLP + paraphrases	0.22	1.4
mid-MLP + paraphrases + distillation	0.36	0.83
attention-top + distillation (control)	0.06	2.9

Placement alone halves damage in 8 of 8 comparisons; paraphrase diversity moves cloze *down* and recall *up* (memorization becoming extraction); distillation from an in-context teacher is the strongest and gentlest write; and the control shows the objective cannot rescue the wrong substrate. Integration then revealed a **third stacked cause**: the safety machinery as designed ($\delta_{\max} = 0.01$) erased the write inside the full pipeline (trained-subset DRA 0.007, three seeds), while a moderate calibration ($\delta_{\max} = 0.1$) preserved both recall and protection (0.189 ± 0.031 at BCP 0.882 ± 0.027 , five seeds, every seed under 1.0). Each cause alone suffices to hold recall at the floor — which is why no single-factor intervention in the diagnosis era ever moved it.

6. The validated system: the four-model matrix

The full validation ran 11–12 August as a staged campaign whose gates, predictions, and stop-loss were committed to the repository before execution: 35 runs, 19.7 GPU-hours, $\approx$ USD \$15.5, zero failures. Every gate was read as written, including the one Llama fails.

6.1 Single cycle, five seeds per model (trained-subset DRA @ BCP)

Model	Recipe	DRA	BCP
Mistral-7B	distil, $\delta_{\max}=0.1$	0.750 ± 0.033	1.21 ± 0.06
Llama-3.1-8B	par.-CE, $\delta_{\max}=0.03$	0.512 ± 0.048	2.09 ± 0.14
Qwen2.5-7B	distil, $\delta_{\max}=0.1$	0.189 ± 0.031	0.88 ± 0.03
Qwen2.5-1.5B	distil, $\delta_{\max}=0.1$	0.105 ± 0.020	1.04 ± 0.02

Every model sits 17–125 $\times$ above the old floor; the strongest result is on the family the recipe never touched during development, which argues against overfitting. Llama is the flagged exception: a pre-registered clip ladder ($0.1 \rightarrow 0.05 \rightarrow 0.03$) reduced damage monotonically ($4.35 \rightarrow 2.14$) without reaching the 1.5 bar — its bleed is not clip-dominated, and it is reported as *not yet tuned*, not as solved.

6.2 Ten cycles, two seeds, against both baselines (DRA cumulative @ BCP at cycle 10)

Model	SLEEP v2-moderate	EWC-only (matched)	Naive LoRA
Qwen2.5-7B	0.173/0.175 @ 1.87/2.24	0.060/0.052 @ 1.53/1.80	0.053/0.060 @ 1.66/1.80
Mistral-7B	0.173/0.168 @ 1.18/1.18	—	0.118/0.160 @ 1.71/1.57
Qwen2.5-1.5B	0.097/0.102 @ 2.35/2.49	—	0.038/0.050 @ 24.5/11.4
Llama-3.1-8B	0.102/0.062 @ 4.33/9.20	0.075/0.072 @ 2.32/1.79	0.082/0.042 @ 2.24/1.67

Three seed-consistent findings. **The design-intent claim is now real:** SLEEP exceeds naive fine-tuning on cumulative recall on all four models, both seeds ($3\times$ on Qwen-7B; on Mistral it wins both axes at once). **The machinery is load-bearing at small scale:** naive training detonates the 1.5B model (BCP 24.5/11.4, seed-random size — the plateau lottery reproduced) while SLEEP holds ≈ 2.4 . **The EWC dissociation:** the matched EWC arm (identical adapter, steps, learning rate; only the penalty differs) posts the best damage numbers with recall at naive’s level — regularisation prevents forgetting; it does not create extraction. SLEEP’s across-seed spread is ≈ 0.005 everywhere; the seed lottery of Section 3 belongs to *unconstrained* arms. On Llama, EWC-only is competitive with the not-yet-tuned SLEEP — reported plainly.

7. Where SLEEP stands, and what we ask

SLEEP set out as a memory system, became a rigorous diagnosis of why the obvious version fails, and is now a working consolidation system with receipts: a frozen model reads once, sleeps, and answers questions from its own weights — validated across three families and two scales, against real baselines, under pre-registered gates. The diagnostic findings stand on their own (the recognition–recall dissociation for injected memory, the near-total proxy-validation failure, the seed-instability of unconstrained baselines), and the meta-lesson generalises: **biological mechanisms transfer; biological anatomy does not** — the substrate must come from the transformer’s own causal structure. Named gaps, carried into future work rather than hidden: cumulative recall decays as early batches fade (interleaved rehearsal — what biological sleep actually does — is the clearest next mechanism); damage leaks $\approx +0.12/\text{cycle}$ in every arm; Llama requires per-family learning-rate work.

The paper has been restructured around this complete arc (31 pages, two movements, nine figures, retitled “*Recognition Without Recall: Diagnosing and Repairing Biologically-Inspired Memory Consolidation in Transformers*”) and is ready for review. We would welcome your judgement on: (i) whether the diagnosis-era first half is correctly weighted against the repair; (ii) target venue and deadline; and (iii) whether the interleaved-rehearsal follow-up should be scoped now or after review.

Respectfully submitted — Aditya Tripathi. All runs, gates, and pre-registrations are in the project repository (logbook/2026-08-11_phase_d_preregistration.md, logbook/2026-08-12_phase_d_full_matrix.md, results under experiments/results/). The proposal (7 July), interim report (27 July), and superseded final (8 August) accompany this document.